

PRIMARY RESEARCH ARTICLE

Positive but variable effects of crop diversification on biodiversity and ecosystem services

Damien Beillouin^{1,2}  | Tamara Ben-Ari^{3,4}  | Eric Malézieux^{1,2}  | Verena Seufert⁵  | David Makowski⁶ ¹CIRAD, UPR HORTSYS, Montpellier, France²HortSys, Univ Montpellier, CIRAD, Montpellier, France³UMR 211, INRAE, AgroParisTech, Université Paris-Saclay, Thiverval-Grignon, France⁴Institut d'écologie et des sciences de l'environnement de Paris (IEES), Campus Pierre et Marie Curie, Paris, France⁵Institute for Environmental Studies (IVM), Vrije Universiteit (VU) Amsterdam, Amsterdam, The Netherlands⁶UMR MIA 518, INRAE, AgroParisTech, Université Paris-Saclay, Paris, France**Correspondence**Damien Beillouin, CIRAD, UPR HORTSYS, F-34398 Montpellier, France.
Email: damien.beillouin@cirad.fr**Funding information**

European Commission, Grant/Award Number: 727482; INRAE-CIRAD; Institute of Convergence CLAND, Grant/Award Number: 16-CONV-0003

Abstract

Ecological theory suggests that biodiversity has a positive and stabilizing effect on the delivery of ecosystem services. Yet, the impacts of increasing the diversity of cultivated crop species or varieties in agroecosystems are still under scrutiny. The available empirical evidence is scattered in scope, agronomic and geographic contexts, and impacts on ecosystem services may depend on the type of diversification strategy used. To robustly assess the effects of crop diversification in agroecosystems, we compiled the results of 95 meta-analyses integrating 5156 experiments conducted over 84 experimental years and representing more than 54,500 paired observations on 120 crop species in 85 countries. Overall, our synthesis of experimental data from across the globe shows that crop diversification enhances not only crop production (median effect +14%) but also the associated biodiversity (+24%, i.e., the biodiversity of non-cultivated plants and animals), and several supporting and regulating ecosystem services including water quality (+51%), pest and disease control (+63%) and soil quality (+11%). However, there was substantial variability in the results for each individual ecosystem service between different diversification strategies such as agroforestry, intercropping, cover crops, crop rotation or variety mixtures. Agroforestry is particularly effective in delivering multiple ecosystem services, that is, water regulation and quality, pest and diseases regulation, associated biodiversity, long-term soil productivity and quality. Variety mixtures, instead, provide the lowest benefits, whereas the other strategies show intermediate results. Our results highlight that while increasing the diversity of cultivated crop species or varieties in agroecosystems represents a very promising strategy for more sustainable land management, contributing to enhanced yields, enhanced biodiversity and ecosystem services, some crop diversification strategies are more effective than others in supporting key ecosystem services.

KEYWORDS

agrobiodiversity, agroforestry, crop rotation, ecosystem services, intercropping, meta-synthesis, research synthesis, rotation

1 | INTRODUCTION

Driven by productivity-oriented policies, agricultural practices throughout the 20th century have converged toward a strong

intensification of agroecosystems (Hazell & Wood, 2008), which was associated with a large decrease in the number of cultivated plant species (Kleijn et al., 2009; Pretty & Bharucha, 2014). These intensive agricultural practices have not only resulted in a

homogenization of agricultural systems but also resulted in agriculture becoming the major contributor of the crossing of several planetary boundaries (Campbell et al., 2017). Various agricultural systems have been proposed as sustainable alternatives to current intensive systems which rely on the use of large amount of external inputs (e.g., fertilizer, pesticides) to ensure adequate food availability while minimizing the environmental impacts of food production (Tamburino et al., 2020), including, for example, organic agriculture, agroecological farming, ecological or sustainable intensification (Cassman & Grassini, 2020; Kremen & Merenlender, 2018; Potts et al., 2017; Pretty & Bharucha, 2014). Most of these alternative systems share the implicit or explicit goal of increasing agrobiodiversity, that is, the diversity of cultivated crop species on-farm and in agricultural landscapes.

Ecological theory suggests that more diverse systems can provide a wider range of ecosystem services and at increased stability over time compared to less diverse systems (Lefcheck et al., 2015; Swift et al., 2004; Tscharnkte et al., 2005). The mechanisms behind the proposed relationship between biodiversity, ecosystem function and the delivery of ecosystem services are manifold, complex and debated. In agroecosystems, these mechanisms are further complicated, since many ecosystem functions are substituted by human labor or by agricultural inputs (Swift et al., 2004).

The growing popularity and promise of crop diversification as a tool for more sustainable land management is evidenced in a fast-growing empirical literature, as well as numerous qualitative (Beillouin et al., 2019; Isbell et al., 2017; Kremen & Miles, 2012; Landis, 2017; Malézieux et al., 2009; Rosa-Schleich et al., 2019; Tscharnkte et al., 2012) and quantitative (Tamburini et al., 2020) syntheses. But when synthesizing this vast body of literature to provide guidance to decision-makers about the promises and shortfalls of these practices, it is crucial to (1) start with a clear definition and

conceptualization of what agricultural diversification means, (2) provide a systematic and global synthesis of the current evidence base while (3) addressing and exploring bias and heterogeneity in the data analyzed, as well as (4) exploring the quality and robustness of the results.

Our objective is hence to provide a rigorous global synthesis of the effects of various crop diversification practices at field scale on biodiversity and ecosystem services. We define crop diversification as any type of agricultural diversification practice that involves increasing the diversity of cultivated annual or perennial crops (including non-harvested crops like cover crops or catch crops) at genetic or species level. To allow for additional insights into the impacts of different types of crop diversification practices, we group crop diversification strategies into five categories: *crop rotations*, *cover crops*, *agroforestry intercropping* and *variety mixtures* (Table 1). To provide a synthesis of the empirical evidence, we compile an exhaustive database of published meta-analyses on the subject and perform a meta-analysis of published meta-analyses, that is, a second-order meta-analysis. This approach has been scarcely applied in agronomy so far, but is already common practice in the medical sciences (Schmidt & Oh, 2013). We also focused on assessing the quality of selected meta-analyses and their redundancy (i.e., how many common primary studies are shared between distinct meta-analyses). The overall summary effect sizes computed in our analysis are weighted according to these criteria (i.e., we perform a weighted second-order meta-analysis, see Section 2). In the following, we present a synthesis of the results of 95 meta-analyses to evaluate the effect of increased crop diversity in comparison to less diverse agroecosystems. Our synthesis allows the identification of strategies that effectively and consistently reduce air, soil and water pollution and increase biodiversity, pest and diseases control and crop production.

Strategies of crop diversification	Definition and details
Agroforestry	The inclusion of woody vegetation, such as trees and shrubs, with crops and/or livestock simultaneously or sequentially on a land management unit. Subcategories include alley cropping, hedgerows, improved fallows, multistrata systems, parkland, shaded perennial-crop systems, silvoarable systems, silvopastures.
Cover crops	The inclusion of plants grown for agronomic or environmental purposes in addition to the main crop on a land management unit. Subcategories include cover crops, grass-strips.
Cultivar mixture	The inclusion of several cultivars of the same species on a land management unit.
Intercropping	The simultaneous or relay cultivation of multiple crops during a significant part of their growth cycle on a land management unit. Subcategories include inter-row grass cover, in flowers/push/pull plants, mixed intercropping, relay intercropping, strip intercropping
Rotation	Recurrent succession of a set of selected crops grown on a same land management unit each season or each year according to a definite plan.

TABLE 1 Short definitions of the crop diversification strategies considered. Extended definitions of strategies and subcategories of strategies can be found in Supplementary Material SII

2 | METHODS

2.1 | Data collection and extraction

We identified gray and published meta-analyses on the impacts of crop diversity on biodiversity and ecosystem services through a systematic search using Web of Science, CAB abstract, Greenfile, Environment Complete Database, Agricola and Google Scholar. The search was performed in May 2020 and updated in November 2020 with the equation: *(meta-analysis OR meta analysis) AND (cropping system OR crop* OR agriculture) AND ((rotation OR Diversification OR intercrop* OR cover crop OR mixture) OR (organic AND (system OR agriculture)) OR (conservation AND (system OR agriculture)) OR no till* OR agroforestry OR agroecology)*. The query was carried out on the article title, abstract and keywords, with no restriction on the date of publication. We also screened the references cited in each selected meta-analysis. This resulted in a total of 1010 publications (857 once duplicates removed—see Prisma flowchart in Supplementary Material SI). Only publications satisfying the following criteria were retained for the analysis: (i) the publication is a meta-analysis reporting the results of a quantitative analysis based on several primary studies, (ii) the meta-analysis focuses on at least one crop diversification strategies defined in Table 1, (iii) the adjacent control plot used in the meta-analysis is less diversified as the treatment (we exclude comparisons to diversified natural ecosystems) and (iv) the meta-analysis presents indicators of precision of the effect sizes (standard error or confidence interval). Studies dealing with pure forestry or wood production were excluded. We did not restrict the selection of studies based on soil–climate conditions, initial level of diversification or other agricultural practices associated with crop diversification. Based on these criteria, 95 meta-analyses were selected.

All quantitative measures of the effects of crop diversification compared to the less diversified reference, that is, effect sizes, were extracted from text, tables or figures using Plot Digitizer (<http://plotdigitizer.sourceforge.net/>). The effect sizes of each meta-analysis were precisely documented (unit, name of the variable, soil, climate, management conditions, etc.). We also retrieved the number of primary studies that were used to calculate the effect sizes, the number of experiments (one primary study can provide several independent estimates), and the confidence intervals or other indicators of dispersion, when available. Effect sizes expressed as standardized mean difference or percent changes were then converted into ratios of the outcomes measured in diversified systems to less diversified systems, to ensure the comparability of the results between the meta-analyses (Borenstein et al., 2009).

Different meta-analyses could be partly based on the same pool of individual experiments. To handle this pseudo-replicated data for our second-order meta-analysis, we retrieved all the references of primary studies of each meta-analysis. We then calculated a redundancy index between each pair of meta-analyses, and included it in our meta-analytical model (see Section 2.2).

The meta-analyses differ in the quality of the methodology used to retrieve the primary study and analyze the data (Beillouin et al.,

2019; Philibert et al., 2012) which can result in biased and misleading results (Nakagawa et al., 2017). To lower the importance of low-quality studies in our second-order meta-analysis, we calculated a quality score, based on 20 criteria analyzing the literature review, statistical analyses and analysis of potential bias of each meta-analysis. We then integrate this score as a proxy of the internal quality of the meta-analyses in our meta-analytical model (see Section 2.2).

The detailed analyses on pseudo-replicated data and quality of the meta-analyses are available in Supplementary Material. All retrieved data are freely available on internet web platform: https://cropdiversification.shinyapps.io/Crop_diversification_2020. Concerning diversification strategies, our dataset is more than five times larger (in terms of number of effect sizes retrieved) than those used in previous papers on the subject (Tamburini et al., 2020).

2.2 | Data analysis

We categorized the retrieved response variables reported in the meta-analyses into main and sub-categories of biodiversity, supporting, regulating, and provisioning ecosystem services. The 12 main categories are agricultural production (40 meta-analyses), soil quality (32 meta-analyses), associated biodiversity (14 meta-analyses), pest and disease control (12 meta-analyses), water quality (11 meta-analyses), water regulation (7 meta-analyses), greenhouse gas (GHG) emissions (7 meta-analyses), input use efficiency (5 meta-analyses), product quality (4 meta-analyses), profitability (3 meta-analyses) and yield stability (1 meta-analysis). A possible wide range of indicators retrieved from the meta-analyses are included in each main category (Supplementary Material SIII). GHG emissions excepted, the higher the value, the higher the “desirability” from the perspective of a farmer or from society as a whole. For example, an increase in associated biodiversity translates into an increase in the abundance, richness or evenness of taxa measured in the diversified systems compared to less diversified ones. Similarly, an increase in water quality, for example, translates in a decrease in nitrate or pesticides concentration in drained water.

We quantified the impacts of crop diversification on each main and sub-categories of biodiversity and ecosystem services. We then detailed the impacts for each crop diversification strategy (e.g., agroforestry, intercropping). The global form of our models is as follows:

$$Y_{ij} = \mu + b_i + \varphi_{ij} + \varepsilon_{ij}, \quad (1)$$

$$\text{with } b_i \sim N(0, \tau^2), \varphi_{ij} \sim N(0, v^2) \text{ and } \varepsilon_{ij} \sim N(0, \sigma_{ij}^2)$$

where Y_{ij} are the j th effect-size of the i th meta-analysis, μ is the mean effect, b_i is the random meta-analysis effect, φ_{ij} is the random effect-size effect within the i th meta-analysis and ε_{ij} is the random estimation errors associated with the j th effect size of the i th meta-analysis. We refer to τ^2 as the between-meta-analysis variance, v^2 as the between effect-size variance and σ_{ij}^2 as the variance of the j th estimated effect size of the i th meta-analysis.

We weighted each meta-analysis by the inverse of the variance of the effect size, as recommended by Marín-Martínez and Sánchez-Meca (2010). We then reduced the weight of the studies of lower quality following Doi et al.'s (2015) methodology (Doi et al., 2015). We considered the non-independence between the effect sizes of different meta-analyses by calculating variance-covariance matrix based on a pseudo-correlation between meta-analyses (Lajeunesse, 2011). The proxy of the correlation between each pair of meta-analyses was estimated as $(2 \times m)/(n_1 + n_2)$, where m is the number of common primary studies between each pair of meta-analyses, and n_1 and n_2 are the total number of primary studies in the two meta-analyses, respectively.

We analyzed the uncertainty and heterogeneity of the estimated effect sizes by calculating (i) the 95% confidence intervals, that is, the range of values that likely contains the mean estimates, (ii) the 95% prediction intervals, that is, the range that likely contains the mean value of a single new observation, a new meta-analysis in our case, (iii) the I² statistics, that is, the percentage of variation across studies that is due to heterogeneity rather than chance (Higgins & Thompson, 2002). We also computed the percentage of variability of each biodiversity and ecosystem services explained by the crop diversification strategy. The added value of *including* crop diversification strategy as moderator in the models was assessed by a likelihood ratio.

The potential publication biases were assessed with funnel plots and Egger tests (Egger et al., 1997). The funnel plots assume that studies with high precision remain close to the average mean effect, and studies with low precision are spread evenly on both sides. The Egger value tests for the asymmetry of the funnel plot. The sensitivity of the results against publication bias was tested with Rosenthal fail-safe number, that is, the number of additional studies with a mean null result necessary to nullify the global estimated effects (see Supplementary Materials SIV and SV). We also provide an estimation of the mean effect considering the missing studies, based on trim-and-fill methodology (Duval & Tweedie, 2000).

We also tested the robustness of our results to the type of model used (with or without random effect), the type of random effects included in the model (only a random meta-analysis effect or a random meta-analysis effect and a random effect-size effect within the i th meta-analysis) and the quality levels of the meta-analyses (model considering quality of the meta-analyses or not).

The parameters of the model were estimated by restricted maximum likelihood. We used log-transformed ratios for performing the statistical analyses. All statistical analyses were conducted with R (version 3.0.2), dplyr (Wickham et al., 2017) for data management, metafor for model analysis (Viechtbauer, 2010) and ggplot2 for data visualization (Wickham, 2016).

3 | RESULTS AND DISCUSSION

We identified 95 meta-analyses synthesizing the effects of increasing crop diversity in agrosystems at the field scale (Figure 1). Our

dataset includes the results of 5156 primary studies equivalent to 54,554 experiments published since 1936, spanning more than 120 crops and 85 countries in Africa, America, Asia, Europe and Oceania (Figure 1).

3.1 | Crop diversification improves biodiversity and ecosystem service delivery

Considered together, the various crop diversification strategies examined improved associated biodiversity (median effect estimate equal to +24%; 95% confidence interval [15%, 33%]), as well as the majority of ecosystem services analyzed (Figure 2), for example resulting in improved water quality (+51%; [3%, 123%]), more efficient pest and disease control (+63%; [31%, 103%]), better soil quality (+11%; [7%, 16%]) and higher production levels (+14%; [8%, 20%]). Our results are thus in line with the positive effects found for highly diverse plant communities in natural ecosystems (Duffy et al., 2017; Loreau & Hector, 2001; Naeem et al., 2009; Tilman et al., 2014) and a previous second-order-meta-analysis for diversified agricultural systems (Tamburini et al., 2020).

The mechanisms behind the positive effect of crop diversification on associated biodiversity, regulating and supporting, as well as provisioning ecosystem services are manifold.

Crop diversification can, for example, provide access to additional resources, or improve the diversity and connectivity of habitats for wildlife (Landis, 2017). Regulatory services like pest and disease control, water or soil quality are, in turn, supported by the higher levels of associated biodiversity, as well as by changes in habitats and resource competition in more diversified systems. Globally, increasing crop diversity has, for example, a strong positive effect on all sub-categories of pest and disease control examined, that is, weed reduction (+60%, [13%, 116%]), pests reduction (+33%, [2%, 74%]), disease control (+41%, [15%, 73%]) and reduction of crop damages (+62%, [37%, 93%]). Increasing crop diversity could alter weed dynamics by limiting available resources (light, water and minerals) for weed species (Weisberger et al., 2019). The increased canopy complexity of diversified systems also provides refuge to natural enemies (Gurr et al., 2017; Langelotto & Denno, 2004), reducing the spread of pests and diseases through microclimatic changes or physical barriers (Finckh et al., 2000). Similarly, crop diversification also improved all measured indicators of soil quality: soil physics, soil chemistry and soil carbon content (see Supplementary Material SVI). The latter indicator is of particular interest to assess the potential for diversified systems to contribute to soil carbon sequestration and climate mitigation although whether and to what degree agricultural practices can result in long-term carbon accumulation in soils is still debated (Amundson & Biardeau, 2018; Poulton et al., 2018).

The higher levels of provisioning services observed in diversified systems could largely be driven by these positive effects on regulatory ecosystem services (Dainese et al., 2019; Gagic et al., 2019; Lundin et al., 2013; Sutter & Albrecht, 2016). Garibaldi et al. (2018) showed that synergistic effects (positive interactions) between

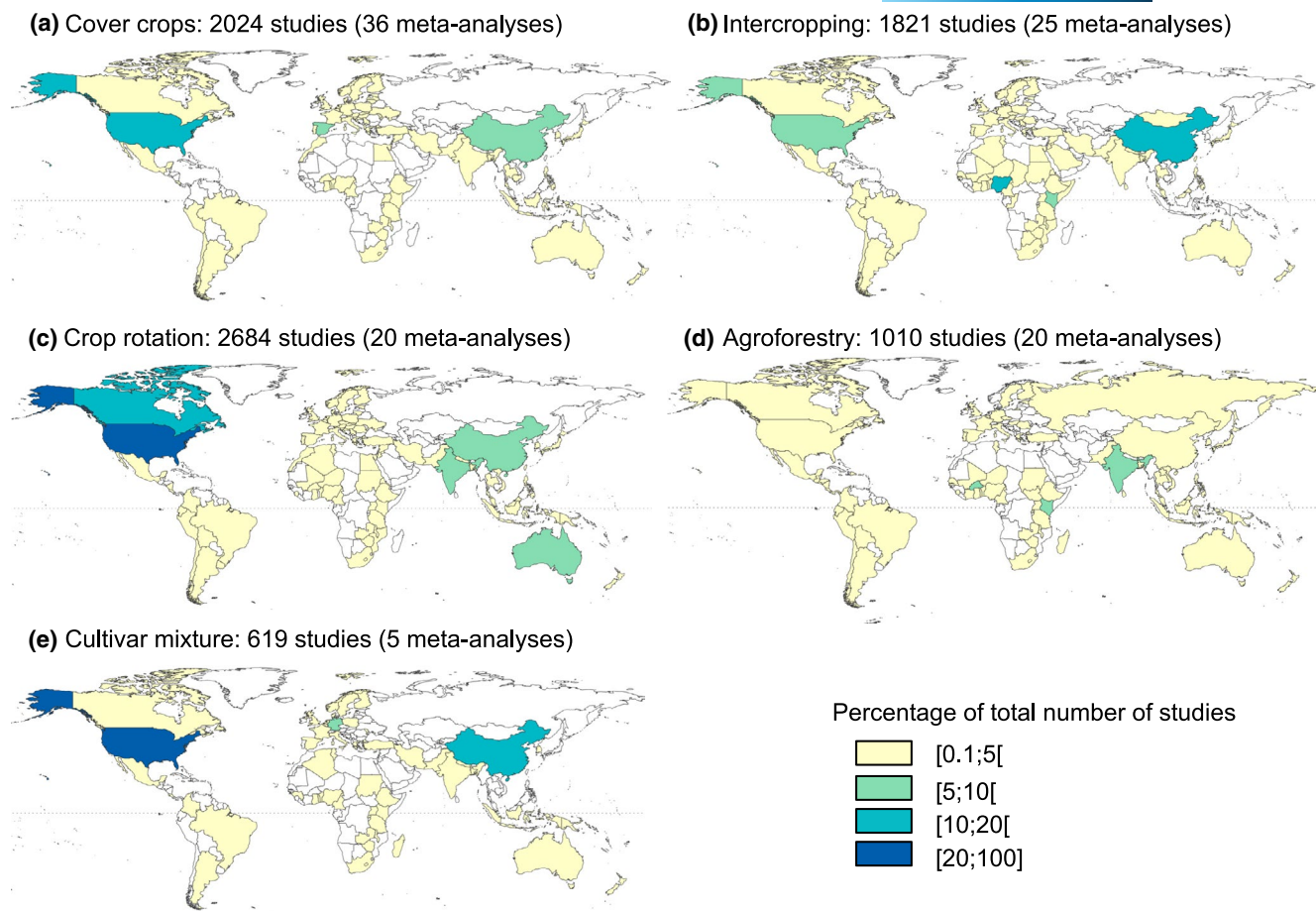


FIGURE 1 Spatial repartition of the primary studies included in the meta-analyses on the five crop diversification strategies included in this paper: Cover crops (a), intercropping (b), crop rotation (c), agroforestry (d), and cultivar mixture (e). The number of meta-analyses from which these primary studies are obtained is noted in brackets. Note that one meta-analysis could study one or multiple crop diversification strategies. For a detailed definition of each crop diversification practice, see Supplementary Material SII

yield and regulatory services generally outnumber negative ones, suggesting that positive impacts of crop diversification on regulating services such as soil quality and pest control lead to the positive impacts on provisioning services observed (Garibaldi et al., 2018).

Although the mean estimates for the regulating and supporting services of water regulation and GHG emissions suggest positive effects, the confidence intervals show that the results remain uncertain (the confidence intervals cross the $X = 0$ line very slightly: +19%, (−1%, 44%) and +25%, (0%, 44%), despite thousands of experiments for these services (1400 and 1300 experiments, respectively). It is also important to note that GHG emissions is the only response variable examined that shows a negative trend—crop diversification practices thus might actually increase GHG emissions, but with results being non-significant due to high uncertainty and/or few syntheses.

In addition, our literature synthesis, although exhaustive, shows that there is currently insufficient aggregated data to precisely evaluate the impacts of crop diversification on input use efficiency, yield stability, product quality and profitability—although the tendency for these outcomes is also positive (i.e., improved input-use efficiency, and product quality, increased profitability). Globally, our

results are robust to the type of statistical model used and to publication bias (see Supplementary Material SVIII).

3.2 | Different crop diversification strategies vary in the magnitude of their positive impacts

Despite these often-clear positive response across all diversification practices, we observe high heterogeneity among the results of different meta-analyses (12 statistics higher than 90% for most ecosystem services—Supplementary Material SVII), which implies that the effects and their magnitude depend on the crop diversification practice and/or bioclimatic conditions. Conducting aggregate analyses of the effects of multiple practices on multiple outcomes (e.g., Tamburini et al., 2020) make thus it difficult to interpret the results to provide accurate guidance for specific policies and farmers. In the following, we show that a substantial part of the variability of the mean effect sizes is due to significant differences in performances between the five crop diversification strategies.

The global positive effect of crop diversification on *associated biodiversity* strongly depends on the type of strategy considered

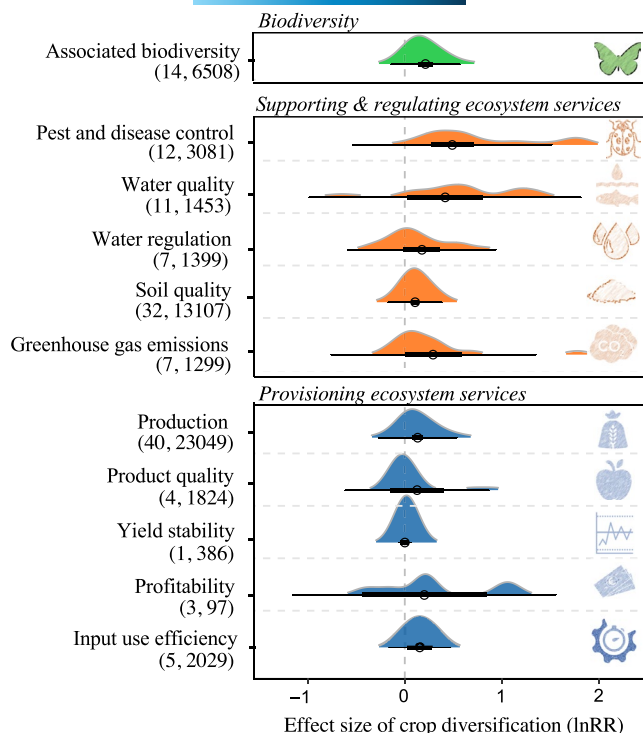


FIGURE 2 Impacts of crop diversification (all strategies) on associated biodiversity, provisioning, supporting and regulating ecosystem services. Points, thick lines and thin lines represent the estimated median summary effect, 95% confidence intervals (CI) and 95% prediction intervals (PI—see Section 2) of the logRatios estimated from a weighted mixed-effect model. CIs indicate the range of values that likely contains the mean estimate of the data analyzed, while PIs indicate the range of values that would likely contain the mean estimate of an additional new meta-analysis. The number of meta-analyses and experiments are displayed between parentheses below each category. Colored shaded areas represent the probability distribution of the effect sizes of the meta-analyses

($p < 0.001$; Figure 3). The biodiversity benefits are the largest for agroforestry (+61%, [26%, 105%]), crop rotation (+37%, [16%, 62%]) and cover crops (+21%, [17%, 25%])—which are all strategies that are often associated with heterogeneous non-crop habitats. Note that we also find substantial differences between different types of agroforestry practices (i.e., +9% of associated biodiversity for silvopasture; +45% for alley cropping; +62% for parklands, +86% for shaded perennial systems, see Figure 3). Intercropping and variety mixtures, on the other hand, have smaller or non-significant effects on associated biodiversity (+7%, [3%, 12%] and +2%, [−12%, 18%], respectively).

The variability of *agricultural production* observed for aggregated results is, also, largely explained by the different crop diversification strategies (67% of the variability of the effect sizes— $p < 0.02$; Figure 3). The two diversification strategies relying on the simultaneous cultivation of different plant species within one field at least during part of their growing cycle (i.e., agroforestry and intercropping) stand out as improving agricultural crop production most strongly (+35%, [12%, 62%]; +22%, [14%, 32%], respectively; note that we do not consider tree production for agroforestry). Including an additional crop in a rotation leads to a 16% median increase ([11%,

21%]) in production levels of the following crops. Cover crops (+6%, [1%, 12%]) and variety mixtures (+2%, [1%, 3%]) promote modest but still significant yield improvements compared to more simplified systems. The remaining variability of the effect sizes within each crop diversification strategy (reflected by the confidence intervals for each of the five strategies) could be caused by the diversity of conditions covered (e.g., a variety of agricultural practices, climates and soil characteristics, crop species or taxa, and initial diversification levels of the control plot). For agroforestry, for example, the different types of systems considered differed in their impact on agricultural production (i.e., −19% production for perennial shaded systems; +3% for hedgerows; +31% for alley cropping, +47% for parklands, see Figure 3). For the other provisioning ecosystem services (i.e. product quality, input use efficiency, yield stability and profitability), the numbers of meta-analyses by crop diversification strategies are rather low, and do not allow a detailed comparison between strategies.

Most *supporting and regulating ecosystem services* also depend on the type of crop diversification strategy implemented. Pest and disease control is, for example, improved by all crop diversification strategies, but with different magnitudes of effects (Figure 4). Our results suggest that the benefits are highest when using cover crops (+125%, [83%, 178%]) while intercropping (+66%, [40%, 98%]) and agroforestry (+59%, [38%, 82%]) still have high but slightly smaller positive impacts. For agroforestry, available evidence shows strong positive effects on pest and disease control for hedgerows (median effect = 84%) and shaded perennial systems (median effect = 40%; Figure 4). The benefits of crop diversification for soil quality are variable across strategies but with low significance ($p < 0.08$ —Figure 4). The benefits are particularly important for agroforestry and intercropping (+19%, [16%, 23%] and +11%, [5%, 18%] respectively). Crop rotation promotes modest soil quality improvements (+5%, CI: 2%, 8%), and variety mixtures have non-significant effects (−5%, [−25%, 18%]). For agroforestry, the median effect is −12% (non-significant) for silvopasture, +13% for hedgerows, +17% for alley cropping and +21% for parklands (Figure 4). Other supporting and regulating ecosystem services such as water regulation (which includes variables such as run-off reduction or soil water infiltration, Supplementary Material) also vary between strategies: Agroforestry leads to significant improvements (+45%, [13%, 87%]) while crop rotation and cover crops effects are non-significant (18%, [−5%, 48%] and 10%, [−10%, 34%], Supplementary Material SIII). On the contrary, no differences are observed among strategies for water quality, partly because of the very large estimated confidence intervals (89%, [20%, 200%] for agroforestry; 87%, [36%, 156%] for intercropping and 61%, [12%, 132%] for cover crop).

3.3 | Climate impacts of crop diversification are variable and context-dependent

Given the importance of agriculture as a contributor to but also a potential solution to anthropogenic climate change (Shukla et al.,

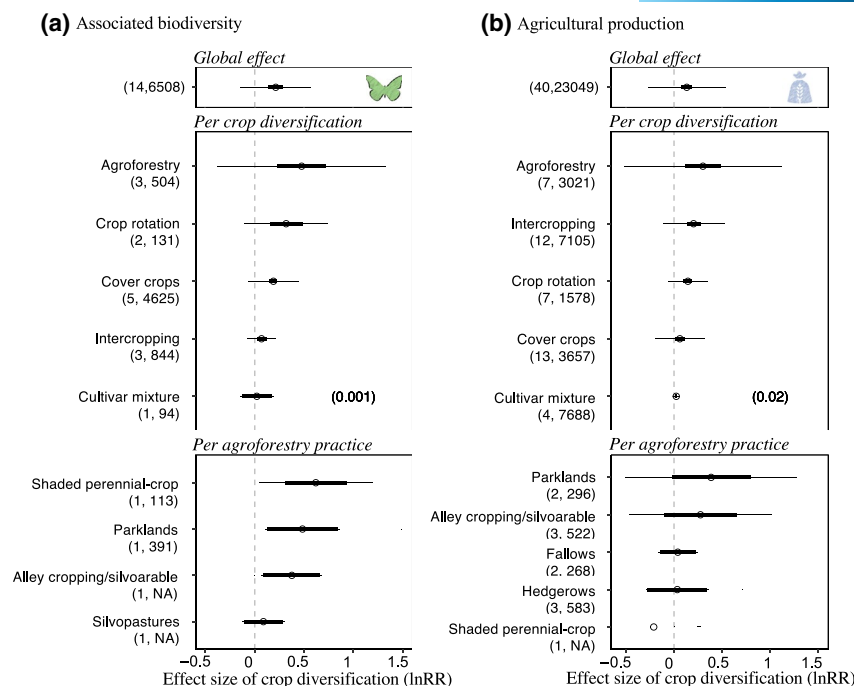


FIGURE 3 Impacts of the five crop diversification strategies on biodiversity (a) and agricultural production (b). Results are presented globally over all strategies, for each of the five crop diversification strategy separately, and for selected agroforestry practices. The numbers of meta-analyses and experiments are displayed between parentheses below each category (NA indicates the absence of reported information in the meta-analyses). Points, thick lines and thin lines represent the estimated median summary effect, 95% confidence intervals (CI) and 95% prediction intervals (PI) of the logRatios estimated from a weighted mixed-effect model. The numbers between parentheses in sub-plots "per crop diversification" represent the *p* values of statistical tests of differences between crop diversification strategies

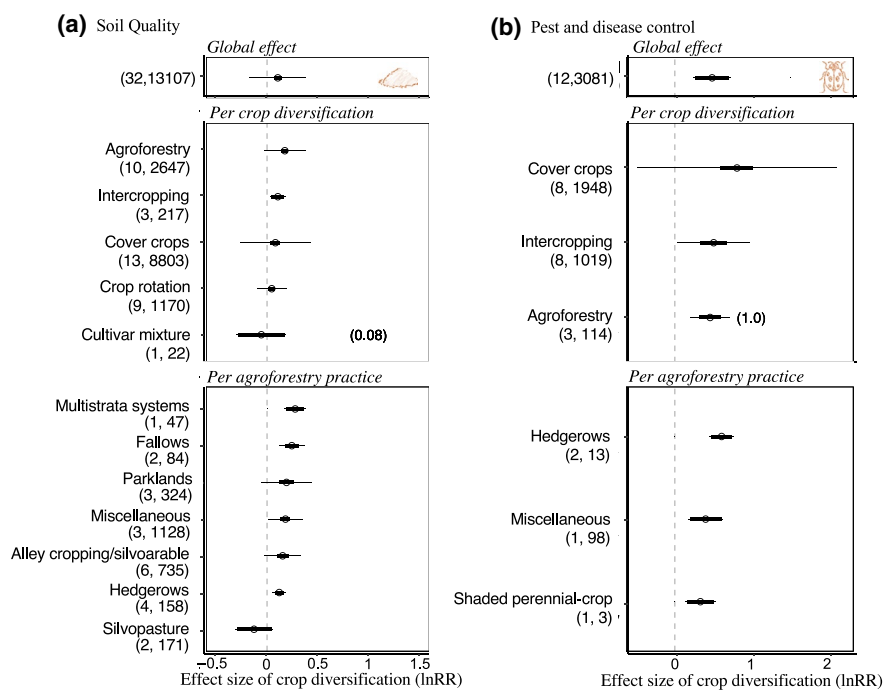


FIGURE 4 Impacts of the five crop diversification strategies on soil quality (a) and pest and disease control (b). Results are presented globally over all strategies, for each of the five crop diversification strategy separately, and selected agroforestry practices. The numbers of meta-analyses and experiments are displayed between parentheses below each category (NA indicates the absence of reported information in the meta-analyses) on the left. Points, thick lines and thin lines represent the estimated median summary effect, 95% confidence intervals (CI) and 95% prediction intervals (PI) of the logRatios estimated from a weighted mixed-effect model. The numbers between parentheses in sub-plots "per crop diversification" represent the *p* values of statistical tests of differences between crop diversification strategies

2019), any sustainable agricultural practice needs to be scrutinized for its climate mitigation potential. We find that agroforestry (+19%, [14%, 24%]), intercropping (+13%, [6%, 10%]) and cover crops (+13%, [10%, 15%]) contribute significantly to increased soil carbon content (see Supplementary Material SVI). On the other hand, both crop rotations and crop variety mixtures have a small and non-significant impact (+3%, [0%, 4%] and -5%, [-24%, 4%], respectively). These different effects of various crop diversification strategies on soil carbon content may be linked with the fact that some strategies such as agroforestry, intercropping and cover crops tend to produce more above- and below-ground litter, which favors the formation and accumulation of soil organic carbon (Wang, et al., 2020). This is in line with other recent studies that have highlighted the promises of agroforestry or cover crops for sequestering carbon in agricultural soils (Bolinder et al., 2020; Chapman et al., 2020). To contribute to soil carbon sequestration and climate mitigation, such practices need not only provide additional biomass and carbon uptake but also (i) not result in a trade-off with yields (which would result in displacement effects), (ii) not result in the increase in other GHGs like nitrous oxide (N_2O) or methane or increased energy use, for example, for fertilizer production and (iii) persist over long periods (Guenet et al., 2021; Powlson et al., 2011). While all diversification practices improve yields (see above), we find based on six meta-analyses that cover crops actually increase GHG emissions (+29%, [+49%, +1]). Experimental evidence suggests that the use of leguminous cover crops, the incorporation of cover crop residues into the soil, as well as the higher availability of mineralizable carbon under cover crop management can result in an increase of direct N_2O emissions compared to non-cover crop management (Abdalla et al., 2019; Mitchell et al., 2013). Yet, a literature synthesis assessing the net GHG balance of cover crops suggests that improved soil carbon stocks, as well as decreased indirect N_2O emissions due to reduced nitrate leaching, could outweigh increased direct N_2O emissions (Abdalla et al., 2019). Note though that carbon sequestration induced by cover crop management is reversible, with a release of the carbon stored over following years (Powlson et al., 2011). The net benefits of cover crops for climate mitigation are thus complex and appear to vary depending on cover crop species, climate or tillage practices (Powlson et al., 2011).

Agroforestry systems, instead, do not only improve yields (Figure 3), but also result in more below-ground soil carbon (Supplementary Materials), as well as more above-ground biomass (Figure 3). Being perennial systems, these effects are also likely to persist longer (Searchinger et al., 2019) and some analyses suggest that they reduce other soil-based GHG emissions like N_2O (Mutuo et al., 2005; which we were not able to assess due to a lack of agroforestry meta-analyses on GHG emissions). These important differences in the performances of different crop diversification practices for soil carbon and climate mitigation highlight the need to unpack such differences instead of treating crop diversification as a monolithic and always sustainable practice.

3.4 | Knowledge gaps and future research agenda

The transition to more sustainable diversified systems is a long and constantly evolving process requiring up-to-date and comprehensive multidisciplinary evaluation. The data gaps that we identified on some important outcomes like yield stability or farm profitability could partly explain the actual low rate of adoption of more diversified strategies in practice. Most of the scientific literature on crop diversification has, to date, focused either on ecological outcomes or on production. Other outcomes are, instead, rarely studied. For example, few results are available on the economic performance of crop diversification strategies (Figure 2), certainly a major factor limiting the rate of adoption of any new agricultural practice (Marra et al., 2003). Restoring ecosystem services or substituting ecosystem functions provided by petrochemical energy will only likely be realized by farmers and land users if there is a direct economic or other benefit to be derived from it (Swift et al., 2004). Similarly, evidence on the stability of the production of diversified crop management is scarce (only one meta-analysis, Figure 2), despite the abundant theoretical literature that suggests that diversified agro-ecosystems might be better able to cope with perturbations (e.g. Newbold et al., 2015; Tamburino et al., 2020). More scientific evidence on the impacts of crop diversity on product quality (e.g. micronutrient content) and input use efficiency—also two variables of interests to farmers and consumers—could also favor broader implementation. Despite a larger database of meta-analyses on crop diversification, we found very few syntheses meeting our criteria analyzing trade-offs between various ecosystem services (contrary to Tamburini et al., 2020). Detailed analyses of the synergies and antagonisms between production and other ecosystem services, such as the one carried out by Lee et al., (2016), have yet to be carried out for the different types of agricultural practices.

Beyond the field scale that constitutes the basis of this study, it is also important to maintain ecosystem services at larger scales, such as farm or landscape, and to promote a higher diversity of land-uses (e.g., Redlich et al., 2018). Most of the ecological processes are strongly dependent on the complex interactions between land use history, biophysical context and human activity at the landscape scale, such as water quantity and quality, pollination, pest regulation or carbon storage (Chase et al., 2018; Li et al., 2009; Swinton et al., 2007). Our analysis does not allow to quantify to what extent such context variables may influence our results. However, in previous studies, it has been shown that levels of natural biodiversity, which varies strongly across different regions (Newbold et al., 2015), may impact crop productions (Pywell et al., 2015).

4 | CONCLUSION

Our results show that increasing the diversity of cultivated crop species or varieties in agroecosystems has a positive but variable impact and depend on the type of crop diversification strategy considered.

Despite the large inherent variability in term of performance of agricultural practices linked with, for example, pedoclimatic conditions, our synthesis allows the identification of strategies that effectively and consistently reduce air, soil and water pollution and increase biodiversity, pest and diseases control and crop production in most situations. Agroforestry strikes out as a particularly promising strategy that is able to substantially increase all the ecosystem services considered in our analysis (and for which sufficient data were available to allow drawing robust conclusions), that is, associated biodiversity, production, water regulation, water quality, pest and diseases control and soil quality. The other diversification strategies also have important effects on specific ecosystem services, for example, cover crops and intercropping on water quality and pests and disease control or crop rotations on associated biodiversity and production. Variety mixtures, instead, provide the lowest benefits for most of the ecosystem services considered.

By precisely weighing the evidence based on its quality, redundancy, geographical coverage and precisely analyzing its heterogeneity and knowledge gaps, we provide a solid basis for policymakers and scientists alike to build future decision-making and future research upon the existing knowledge base. It can contribute to help the Food agricultural Organization (FAO) monitoring and orienting policies towards Sustainable Development Goals or provide support for the new European “Green Deal” as a starting point for regional debates on the effectiveness of the various strategies analyzed globally (The European Green Deal, 2019). Our study can also encourage to fill the knowledge gaps on some outcomes, such as yield stability and farm profitability.

ACKNOWLEDGEMENTS

This work was produced within the framework of the European project “Diversification through Rotation, Intercropping, Multiple Cropping, Promoted with Actors and value-Chains towards Sustainability” (DiverIMPACTS), funded by the European Commission under Grant Agreement number 727482. It was also supported by the INRAE-CIRAD metaprogram GloFoods and by the Institute of Convergence CLAND (16-CONV-0003). We are grateful to Mathilde Duvallet for her contribution to the database.

CONFLICT OF INTERESTS

The authors declare no competing interests.

AUTHOR CONTRIBUTION

DM, TBA and DB participated in the design and the coordination of the study. DB performs the data collection, with the help of DM and TBA. DB performed the statistical analyses with the help of DM, and designed the figures. DB and TBA wrote the draft, all authors provided critical feedback and contributed significantly to the writing.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available on an internet web platform https://cropdiversification.shinyapps.io/Crop_diversification_2020.

ORCID

Damien Beillouin  <https://orcid.org/0000-0003-2014-3482>

Tamara Ben-Ari  <https://orcid.org/0000-0001-7157-7905>

Eric Malézieux  <https://orcid.org/0000-0002-5706-9610>

Verena Seufert  <https://orcid.org/0000-0003-2975-5856>

David Makowski  <https://orcid.org/0000-0001-6385-3703>

REFERENCES

- Abdalla, M., Hastings, A., Cheng, K., Yue, Q., Chadwick, D., Espenberg, M., Truu, J., Rees, R. M., & Smith, P. (2019). A critical review of the impacts of cover crops on nitrogen leaching, net greenhouse gas balance and crop productivity. *Global Change Biology*, 25(8), 2530–2543. <https://doi.org/10.1111/gcb.14644>
- Amundson, R., & Biardeau, L. (2018). Opinion: Soil carbon sequestration is an elusive climate mitigation tool. *Proceedings of the National Academy of Sciences of the United States of America*, 115(46), 11652–11656. <https://doi.org/10.1073/pnas.1815901115>
- Beillouin, D., Ben-Ari, T., & Makowski, D. (2019). Evidence map of crop diversification strategies at the global scale. *Environmental Research Letters*, 14(12), 123001. <https://doi.org/10.1088/1748-9326/ab4449>
- Bolinder, M. A., Crotty, F., Elsen, A., Frac, M., Kismányoky, T., Lipiec, J., Tits, M., Tóth, Z., & Kätterer, T. (2020). The effect of crop residues, cover crops, manures and nitrogen fertilization on soil organic carbon changes in agroecosystems: A synthesis of reviews. *Mitigation and Adaptation Strategies for Global Change*, 25(6), 929–952. <https://doi.org/10.1007/s11027-020-09916-3>
- Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. John Wiley & Sons, Ltd. <https://doi.org/10.1002/9780470743386>
- Campbell, B. M., Beare, D. J., Bennett, E. M., Hall-Spencer, J. M., Ingram, J. S. I., Jaramillo, F., Ortiz, R., Ramankutty, N., Sayer, J. A., & Shindell, D. (2017). Agriculture production as a major driver of the Earth system exceeding planetary boundaries. *Ecology and Society*, 22(4), art8. <https://doi.org/10.5751/ES-09595-220408>
- Cassman, K. G., & Grassini, P. (2020). A global perspective on sustainable intensification research. *Nature Sustainability*, 3(4), 262–268. <https://doi.org/10.1038/s41893-020-0507-8>
- Chapman, M., Walker, W. S., Cook-Patton, S. C., Ellis, P. W., Farina, M., Griscom, B. W., & Baccini, A. (2020). Large climate mitigation potential from adding trees to agricultural lands. *Global Change Biology*, 26(8), 4357–4365. <https://doi.org/10.1111/gcb.15121>
- Chase, J. M., McGill, B. J., McGlinn, D. J., May, F., Blowes, S. A., Xiao, X., Knight, T. M., Purschke, O., & Gotelli, N. J. (2018). Embracing scale-dependence to achieve a deeper understanding of biodiversity and its change across communities. *Ecology Letters*, 21(11), 1737–1751. <https://doi.org/10.1111/ele.13151>
- Dainese, M., Martin, E. A., Aizen, M. A., Albrecht, M., Bartomeus, I., Bommarco, R., Carvalheiro, L. G., Chaplin-Kramer, R., Gagic, V., Garibaldi, L. A., Ghazoul, J., Grab, H., Jonsson, M., Karp, D. S., Kennedy, C. M., Kleijn, D., Kremen, C., Landis, D. A., Letourneau, D. K., ... Steffan-Dewenter, I. (2019). A global synthesis reveals biodiversity-mediated benefits for crop production. *Science Advances*, 5(10), eaax0121. <https://doi.org/10.1126/sciadv.aax0121>
- Doi, S. A. R., Barendregt, J. J., Khan, S., Thalib, L., & Williams, G. M. (2015). Advances in the meta-analysis of heterogeneous clinical trials II: The quality effects model. *Contemporary Clinical Trials*, 45, 123–129. <https://doi.org/10.1016/j.cct.2015.05.010>
- Duffy, J. E., Godwin, C. M., & Cardinale, B. J. (2017). Biodiversity effects in the wild are common and as strong as key drivers of productivity. *Nature*, 549(7671), 261–264. <https://doi.org/10.1038/nature23886>

- Duval, S., & Tweedie, R. (2000). A nonparametric "trim and fill" method of accounting for publication bias in meta-analysis. *Journal of the American Statistical Association*, 95(449), 89–98. <https://doi.org/10.1080/01621459.2000.10473905>
- Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ*, 315(7109), 629–634. <https://doi.org/10.1136/bmj.315.7109.629>
- Finckh, M. R., Gacek, E. S., Goyeau, H., Lannou, C., Merz, U., Mundt, C. C., Munk, L., Nadziak, J., Newton, A. C., de Vallavieille-Pope, C., & Wolfe, M. S. (2000). Cereal variety and species mixtures in practice, with emphasis on disease resistance. *Agronomie*, 20(7), 813–837. <https://doi.org/10.1051/agro:2000177>
- Gagic, V., Marcora, A., & Howie, L. (2019). Additive and interactive effects of pollination and biological pest control on crop yield. *Journal of Applied Ecology*, 56(11), 2528–2535. <https://doi.org/10.1111/1365-2664.13482>
- Garibaldi, L. A., Andersson, G. K. S., Requier, F., Fijen, T. P. M., Hipólito, J., Kleijn, D., Pérez-Méndez, N., & Rollin, O. (2018). Complementarity and synergisms among ecosystem services supporting crop yield. *Global Food Security*, 17, 38–47. <https://doi.org/10.1016/j.gfs.2018.03.006>
- Guenet, B., Gabrielle, B., Chenu, C., Arrouays, D., Balesdent, J., Bernoux, M., Bruni, E., Caliman, J.-P., Cardinael, R., Chen, S., Ciais, P., Desbois, D., Fouche, J., Frank, S., Henault, C., Lugato, E., Naipal, V., Nesme, T., Obersteiner, M., ... Zhou, F. (2021). Can N₂O emissions offset the benefits from soil organic carbon storage? *Global Change Biology*, 27(2), 237–256. <https://doi.org/10.1111/gcb.15342>
- Gurr, G. M., Wratten, S. D., Landis, D. A., & You, M. (2017). Habitat management to suppress pest populations: Progress and prospects. *Annual Review of Entomology*, 62(1), 91–109. <https://doi.org/10.1146/annurev-ento-031616-035050>
- Hadley Wickham, R. F., Henry, L., & Müller, K. (2017). Dplyr: A grammar of data manipulation. R package version 0.7.4.
- Hazell, P., & Wood, S. (2008). Drivers of change in global agriculture. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 363(1491), 495–515. <https://doi.org/10.1098/rstb.2007.2166>
- Higgins, J. P., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, 21(11), 1539–1558. <https://doi.org/10.1002/sim.1186>
- Isbell, F., Adler, P. R., Eisenhauer, N., Fornara, D., Kimmel, K., Kremen, C., Letourneau, D. K., Liebman, M., Polley, H. W., Quijas, S., & Scherer-Lorenzen, M. (2017). Benefits of increasing plant diversity in sustainable agroecosystems. *Journal of Ecology*, 105(4), 871–879. <https://doi.org/10.1111/1365-2745.12789>
- Kleijn, D., Kohler, F., Báldi, A., Batáry, P., Concepción, E. D., Clough, Y., Díaz, M., Gabriel, D., Holzschuh, A., Knop, E., Kovács, A., Marshall, E. J. P., Tscharntke, T., & Verhulst, J. (2009). On the relationship between farmland biodiversity and land-use intensity in Europe. *Proceedings of the Royal Society B: Biological Sciences*, 276(1658), 903–909. <https://doi.org/10.1098/rspb.2008.1509>
- Kremen, C., & Merenlender, A. M. (2018). Landscapes that work for biodiversity and people. *Science*, 362(6412), eaau6020. <https://doi.org/10.1126/science.aau6020>
- Kremen, C., & Miles, A. (2012). Ecosystem services in biologically diversified versus conventional farming systems: Benefits, externalities, and trade-offs. *Ecology and Society*, 17(4), art40. <https://doi.org/10.5751/ES-05035-170440>
- Lajeunesse, M. J. (2011). On the meta-analysis of response ratios for studies with correlated and multi-group designs. *Ecology*, 92(11), 2049–2055. <https://doi.org/10.1890/11-0423.1>
- Landis, D. A. (2017). Designing agricultural landscapes for biodiversity-based ecosystem services. *Basic and Applied Ecology*, 18, 1–12. <https://doi.org/10.1016/j.baae.2016.07.005>
- Langellotto, G. A., & Denno, R. F. (2004). Responses of invertebrate natural enemies to complex-structured habitats: A meta-analytical synthesis. *Oecologia*, 139(1), 1–10. <https://doi.org/10.1007/s00442-004-1497-3>
- Lee, H., & Lautenbach, S. (2016). A quantitative review of relationships between ecosystem services. *Ecological Indicators*, 66, 340–351.
- Lefcheck, J. S., Byrnes, J. E., Isbell, F., Gamfeldt, L., Griffin, J. N., Eisenhauer, N., Hensel, M. J., Hector, A., Cardinale, B. J., & Duffy, J. E. (2015). Biodiversity enhances ecosystem multifunctionality across trophic levels and habitats. *Nature Communications*, 6(1), 1–7. <https://doi.org/10.1038/ncomms7936>
- Li, C., He, X., Zhu, S., Zhou, H., Wang, Y., Li, Y., Yang, J., Fan, J., Yang, J., Wang, G., Long, Y., Xu, J., Tang, Y., Zhao, G., Yang, J., Liu, L., Sun, Y., Xie, Y., Wang, H., & Zhu, Y. (2009). Crop diversity for yield increase. *PLoS ONE*, 4(11), e8049. <https://doi.org/10.1371/journal.pone.0008049>
- Loreau, M., & Hector, A. (2001). Partitioning selection and complementarity in biodiversity experiments. *Nature*, 412(6842), 72–76. <https://doi.org/10.1038/35083573>
- Lundin, O., Smith, H. G., Rundlöf, M., & Bommarco, R. (2013). When ecosystem services interact: Crop pollination benefits depend on the level of pest control. *Proceedings of the Royal Society B: Biological Sciences*, 280(1753), 20122243. <https://doi.org/10.1098/rspb.2012.2243>
- Malézieux, E., Crozat, Y., Dupraz, C., Laurans, M., Makowski, D., Ozier-Lafontaine, H., Rapidel, B., Tourdonnet, S., & Valantin-Morison, M. (2009). Mixing plant species in cropping systems: Concepts, tools and models. A review. *Agronomy for Sustainable Development*, 29(1), 43–62. <https://doi.org/10.1051/agro:2007057>
- Marín-Martínez, F., & Sánchez-Meca, J. (2010). Weighting by inverse variance or by sample size in random-effects meta-analysis. *Educational and Psychological Measurement*, 70(1), 56–73. <https://doi.org/10.1177/0013164409344534>
- Marra, M., Pannell, D. J., & Ghadim, A. A. (2003). The economics of risk, uncertainty and learning in the adoption of new agricultural technologies: Where are we on the learning curve? *Agricultural Systems*, 75(2–3), 215–234. [https://doi.org/10.1016/S0308-521X\(02\)00066-5](https://doi.org/10.1016/S0308-521X(02)00066-5)
- Mitchell, D. C., Castellano, M. J., Sawyer, J. E., & Pantoja, J. (2013). Cover crop effects on nitrous oxide emissions: Role of mineralizable carbon. *Soil Science Society of America Journal*, 77(5), 1765–1773. <https://doi.org/10.2136/sssaj2013.02.0074>
- Mutuo, P. K., Cadisch, G., Albrecht, A., Palm, C., & Verchot, L. (2005). Potential of agroforestry for carbon sequestration and mitigation of greenhouse gas emissions from soils in the tropics. *Nutrient Cycling in Agroecosystems*, 71(1), 43–54. <https://doi.org/10.1007/s10705-004-5285-6>
- Naeem, S., Bunker, D. E., Hector, A., Loreau, M., & Perrings, C. (2009). *Biodiversity, ecosystem functioning, and human wellbeing: An ecological and economic perspective*. Oxford University Press.
- Nakagawa, S., Noble, D. W., Senior, A. M., & Lagisz, M. (2017). Meta-evaluation of meta-analysis: Ten appraisal questions for biologists. *BMC Biology*, 15(1), 1–14. <https://doi.org/10.1186/s12915-017-0357-7>
- Newbold, T., Hudson, L. N., Hill, S. L. L., Contu, S., Lysenko, I., Senior, R. A., Börger, L., Bennett, D. J., Choimes, A., Collen, B., Day, J., De Palma, A., Díaz, S., Echeverria-Londoño, S., Edgar, M. J., Feldman, A., Garon, M., Harrison, M. L. K., Alhusseini, T., ... Purvis, A. (2015). Global effects of land use on local terrestrial biodiversity. *Nature*, 520(7545), 45–50. <https://doi.org/10.1038/nature14324>
- Philibert, A., Loyce, C., & Makowski, D. (2012). Assessment of the quality of meta-analysis in agronomy. *Agriculture, Ecosystems & Environment*, 148, 72–82. <https://doi.org/10.1016/j.agee.2011.12.003>
- Potts, S. G., Imperatriz-Fonseca, V. L., & Ngo, H. T. (2017). *IPBES the assessment report of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services on pollinators, pollination and food production* (p. 552). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services.

- Poulton, P., Johnston, J., Macdonald, A., White, R., & Powlson, D. (2018). Major limitations to achieving "4 per 1000" increases in soil organic carbon stock in temperate regions: Evidence from long-term experiments at Rothamsted Research, United Kingdom. *Global Change Biology*, 24(6), 2563–2584. <https://doi.org/10.1111/gcb.14066>
- Powlson, D. S., Whitmore, A. P., & Goulding, K. W. (2011). Soil carbon sequestration to mitigate climate change: A critical re-examination to identify the true and the false. *European Journal of Soil Science*, 62(1), 42–55. <https://doi.org/10.1111/j.1365-2389.2010.01342.x>
- Pretty, J., & Bharucha, Z. P. (2014). Sustainable intensification in agricultural systems. *Annals of Botany*, 114(8), 1571–1596. <https://doi.org/10.1093/aob/mcu205>
- Pywell, R. F., Heard, M. S., Woodcock, B. A., Hinsley, S., Ridding, L., Nowakowski, M., & Bullock, J. M. (2015). Wildlife-friendly farming increases crop yield: Evidence for ecological intensification. *Proceedings of the Royal Society B: Biological Sciences*, 282(1816), 20151740. <https://doi.org/10.1098/rspb.2015.1740>
- Redlich, S., Martin, E. A., & Steffan-Dewenter, I. (2018). Landscape-level crop diversity benefits biological pest control. *Journal of Applied Ecology*, 55(5), 2419–2428. <https://doi.org/10.1111/1365-2664.13126>
- Rosa-Schleich, J., Loos, J., Mußhoff, O., & Tscharnkte, T. (2019). Ecological-economic trade-offs of Diversified Farming Systems – A review. *Ecological Economics*, 160, 251–263. <https://doi.org/10.1016/j.ecolecon.2019.03.002>
- Schmidt, F. L., & Oh, I.-S. (2013). Methods for second order meta-analysis and illustrative applications. *Organizational Behavior and Human Decision Processes*, 121(2), 204–218. <https://doi.org/10.1016/j.obhdp.2013.03.002>
- Searchinger, T., Waite, R., Hanson, C., Ranganathan, J., Dumas, P., Matthews, E., & Klirs, C. (2019). *Creating a sustainable food future: A menu of solutions to feed nearly 10 billion people by 2050. Final report.* WRI.
- Shukla, P., Skea, J., Calvo Buendia, E., Masson-Delmotte, V., Pörtner, H., Roberts, D., Zhai, P., Slade, R., Connors, S., & Van Diemen, R. (2019). *IPCC, 2019: Climate change and land: An IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems.* <https://www.ipcc.ch/srccl/download/>
- Sutter, L., & Albrecht, M. (2016). Synergistic interactions of ecosystem services: Florivorous pest control boosts crop yield increase through insect pollination. *Proceedings of the Royal Society B: Biological Sciences*, 283(1824), 20152529. <https://doi.org/10.1098/rspb.2015.2529>
- Swift, M. J., Izac, A.-M.-N., & van Noordwijk, M. (2004). Biodiversity and ecosystem services in agricultural landscapes—Are we asking the right questions? *Agriculture, Ecosystems & Environment*, 104(1), 113–134. <https://doi.org/10.1016/j.agee.2004.01.013>
- Swinton, S. M., Lupi, F., Robertson, G. P., & Hamilton, S. K. (2007). *Ecosystem services and agriculture: Cultivating agricultural ecosystems for diverse benefits.* Elsevier. <https://doi.org/10.1016/j.ecolecon.2007.09.020>
- Tamburini, G., Bommarco, R., Wanger, T. C., Kremen, C., van der Heijden, M. G., Liebman, M., & Hallin, S. (2020). Agricultural diversification promotes multiple ecosystem services without compromising yield. *Science Advances*, 6(45), eaba1715. <https://doi.org/10.1126/sciadv.aba1715>
- Tamburino, L., Bravo, G., Clough, Y., & Nicholas, K. A. (2020). From population to production: 50 years of scientific literature on how to feed the world. *Global Food Security*, 24, 100346. <https://doi.org/10.1016/j.gfs.2019.100346>
- The European Green Deal. COM 2019(640) final. (2019). <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52019DC0640>
- Tilman, D., Isbell, F., & Cowles, J. M. (2014). Biodiversity and ecosystem functioning. *Annual Review of Ecology, Evolution, and Systematics*, 45, 471–493. <https://doi.org/10.1146/annurev-ecolsys-120213-091917>
- Tscharnkte, T., Clough, Y., Wanger, T. C., Jackson, L., Motzke, I., Perfecto, I., Vandermeer, J., & Whitbread, A. (2012). Global food security, biodiversity conservation and the future of agricultural intensification. *Biological Conservation*, 151(1), 53–59. <https://doi.org/10.1016/j.biocon.2012.01.068>
- Tscharnkte, T., Klein, A. M., Kruess, A., Steffan-Dewenter, I., & Thies, C. (2005). Landscape perspectives on agricultural intensification and biodiversity—Ecosystem service management. *Ecology Letters*, 8(8), 857–874. <https://doi.org/10.1111/j.1461-0248.2005.00782.x>
- Viechtbauer, W. (2010). Conducting meta-analyses in R with the **metafor** package. *Journal of Statistical Software*, 36(3), <https://doi.org/10.18637/jss.v036.i03>
- Wang, Y., Xu, Y., Pei, J., Li, M., Shan, T., Zhang, W., & Wang, J. (2020). Below ground residues were more conducive to soil organic carbon accumulation than above ground ones. *Applied Soil Ecology*, 148, 103509. <https://doi.org/10.1016/j.apsoil.2020.103509>
- Weisberger, D., Nichols, V., & Liebman, M. (2019). Does diversifying crop rotations suppress weeds? A meta-analysis. *PLoS ONE*, 14(7), e0219847. <https://doi.org/10.1371/journal.pone.0219847>
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis.* Springer-Verlag. <https://ggplot2.tidyverse.org>

META-ANALYSES INCLUDED IN OUR STUDY

- Abdalla, M., Hastings, A., Cheng, K., Yue, Q., Chadwick, D., Espenberg, M., Truu, J., Rees, R. M., & Smith, P. (2019). A critical review of the impacts of cover crops on nitrogen leaching, net greenhouse gas balance and crop productivity. *Global Change Biology*, 25(8), 2530–2543. <https://doi.org/10.1111/gcb.14644>
- Aguilera, E., Lassaletta, L., Gattinger, A., & Gimeno, B. S. (2013). Managing soil carbon for climate change mitigation and adaptation in Mediterranean cropping systems. A meta-analysis. *Agriculture Ecosystems & Environment*, 168, 25–36. <https://doi.org/10.1016/j.agee.2013.02.003>
- Alvarez, R., Steinbach, H. S., & De Paepe, J. L. (2017). Cover crop effects on soils and subsequent crops in the pampas: A meta-analysis. *Soil and Tillage Research*, 170, 53–65. <https://doi.org/10.1016/j.still.2017.03.005>
- Andres, C., Gattinger, A., Dzahini-Obiatey, H. K., Blaser, W. J., Offei, S. K., & Six, J. (2017). Combatting cocoa swollen shoot virus disease. What do we know? *Crop Protection*, 98, 76–84. <https://doi.org/10.1016/j.cropro.2017.03.010>
- Ashworth, A. J., Toler, H. D., Allen, F. L., & Augé, R. M. (2018). Global meta-analysis reveals agro-grassland productivity varies based on species diversity over time. *PLoS ONE*, 13(7), e0200274. <https://doi.org/10.1371/journal.pone.0200274>
- Bai, X., Huang, Y., Ren, W., Coyne, M., Jacinthe, P. A., Tao, B., Hui, D., Yang, J., & Matocha, C. (2019). Responses of soil carbon sequestration to climate-smart agriculture practices: A meta-analysis. *Global Change Biology*, 25(8), 2591–2606. <https://doi.org/10.1111/gcb.14658>
- Bai, Z., Caspari, T., Gonzalez, M. R., Batjes, N. H., Mäder, P., Bünemann, E. K., de Goede, R., Brussaard, L., Xu, M., Ferreira, C. S. S., Reintam, E., Fan, H., Mihelič, R., Glavan, M., & Tóth, Z. (2018). Effects of agricultural management practices on soil quality: A review of long-term experiments for Europe and China. *Agriculture, Ecosystems & Environment*, 265, 1–7. <https://doi.org/10.1016/j.agee.2018.05.028>
- Basche, A. D., Miguez, F. E., Kaspar, T. C., & Castellano, M. J. (2014). Do cover crops increase or decrease nitrous oxide emissions? A meta-analysis. *Journal of Soil and Water Conservation*, 69, 471–482. <https://doi.org/10.2489/jswc.69.6.471>
- Basche, A. D., & DeLonge, M. S. (2019). Comparing infiltration rates in soils managed with conventional and alternative farming methods: A meta-analysis. *BioRxiv*, 603696.
- Bayala, J., Kalinganire, A., Sileshi, G. W., & Tondoh, J. E. (2018). Soil organic carbon and nitrogen in agroforestry systems in sub-Saharan Africa: A review. *Improving the profitability, sustainability and efficiency of nutrients through site specific fertilizer recommendations in west Africa agro-ecosystems* (pp. 51–61). Springer International Publishing.
- Laura, V. V., Bert, R., Steven, B., Pieter, D. F., Victoria, N., Paul, P., & Kris, V. (2017). Ecosystem service delivery of agri-environment measures: A synthesis for hedgerows and grass strips on arable land. *Agriculture, Ecosystems & Environment*, 244, 32–51. <https://doi.org/10.1016/j.agee.2017.04.015>
- Borg, J., Kiær, L. P., Lecarpentier, C., Goldringer, I., Gauffreteau, A., Saint-Jean, S., Barot, S., & Enjalbert, J. (2018). Unfolding the potential of wheat cultivar mixtures: A meta-analysis perspective and identification of knowledge gaps. *Field Crops Research*, 221, 298–313. <https://doi.org/10.1016/j.fcr.2017.09.006>

- Bowles, T. M., Jackson, L. E., Loeher, M., & Cavagnaro, T. R. (2017). Ecological intensification and arbuscular mycorrhizas: A meta-analysis of tillage and cover crop effects. *Journal of Applied Ecology*, 54(6), 1785–1793. <https://doi.org/10.1111/1365-2664.12815>
- Cernay, C., Makowski, D., & Pelzer, E. (2017). Preceding cultivation of grain legumes increases cereal yields under low nitrogen input conditions. *Environmental Chemistry Letters*, 16(2), 631–636. <https://doi.org/10.1007/s10311-017-0698-z>
- Chastaing, G. (2009). Meta-analysis on grain yield effects of cereals-legumes intercropping. Second Years Training Report.
- Chen, C., Chen, H. Y., Chen, X., & Huang, Z. (2019). Meta-analysis shows positive effects of plant diversity on microbial biomass and respiration. *Nature Communications*, 10(1), 1–10. <https://doi.org/10.1038/s41467-019-09258-y>
- Chen, X., Chen, H. Y., Chen, C., Ma, Z., Searle, E. B., Yu, Z., & Huang, Z. (2020). Effects of plant diversity on soil carbon in diverse ecosystems: A global meta-analysis. *Biological Reviews*, 95(1), 167–183. <https://doi.org/10.1111/brv.12554>
- Congreves, K. A., Smith, J. M., Nemeth, D. D., Hooker, D. C., & Van Eerd, L. L. (2014). Soil organic carbon and land use. Processes and potential in Ontario long-term agro-ecosystem research sites. *Canadian Journal of Soil Science*, 94, 317–336. <https://doi.org/10.4141/cjss2013-094>
- Crowder, D. W., & Reganold, J. P. (2015). Financial competitiveness of organic agriculture on a global scale. *Proceedings of the National Academy of Sciences of the United States of America*, 112, 7611–7616. <https://doi.org/10.1073/pnas.1423674112>
- Daryanto, S., Fu, B., Zhao, W., Wang, S., Jacinthe, P. A., & Wang, L. (2020). Ecosystem service provision of grain legume and cereal intercropping in Africa. *Agricultural Systems*, 178, 102761. <https://doi.org/10.1016/j.agsy.2019.102761>
- De Beenhouwer, M., Aerts, R., & Honnay, O. (2013). A global meta-analysis of the biodiversity and ecosystem service benefits of coffee and cacao agroforestry. *Agriculture, Ecosystems & Environment*, 175, 1–7. <https://doi.org/10.1016/j.agee.2013.05.003>
- Decock, C. (2014). Mitigating nitrous oxide emissions from corn cropping systems in the Midwestern US: Potential and data gaps. *Environmental Science & Technology*, 48(8), 4247–4256. <https://doi.org/10.1021/es4055324>
- Francaviglia, R., Álvaro-Fuentes, J., Di Bene, C., Gai, L., Regina, K., & Turtola, E. (2020). Diversification and management practices in selected European regions. A data analysis of arable crops production. *Agronomy*, 10(2), 297. <https://doi.org/10.3390/agronomy10020297>
- Gardner, J. B., & Drinkwater, L. E. (2009). The fate of nitrogen in grain cropping systems. A meta-analysis of N-15 field experiments. *Ecological Applications*, 19, 2167–2184.
- Gu, J., Nie, H., Guo, H., Xu, H., & Gunnathorn, T. (2019). Nitrous oxide emissions from fruit orchards: A review. *Atmospheric Environment*, 201, 166–172. <https://doi.org/10.1016/j.atmosenv.2018.12.046>
- Han, Z., Walter, M. T., & Drinkwater, L. E. (2017). N₂O emissions from grain cropping systems. A meta-analysis of the impacts of fertilizer-based and ecologically-based nutrient management strategies. *Nutrient Cycling in Agroecosystems*, 107, 335–355. <https://doi.org/10.1007/s10705-017-9836-z>
- Himmelstein, J., Ares, A., Gallagher, D., & Myers, J. (2017). A meta-analysis of intercropping in Africa. Impacts on crop yield, farmer income, and integrated pest management effects. *International Journal of Agricultural Sustainability*, 15, 1–10. <https://doi.org/10.1080/14735903.2016.1242332>
- Iverson, A. L., Marin, L. E., Ennis, K. K., Gonther, D. J., Connor-Barrie, B. T., Remfert, J. L., Cardinale, B. J., & Perfecto, I. (2014). Do polycultures promote win-wins or trade-offs in agricultural ecosystem services? A meta-analysis. *Journal of Applied Ecology*, 51, 1593–1602.
- Jian, J., Du, X., Reiter, M. S., & Stewart, R. D. (2020). A meta-analysis of global cropland soil carbon changes due to cover cropping. *Soil Biology and Biochemistry*, 107735. <https://doi.org/10.1016/j.soilbio.2020.107735>
- Jian, J., Lester, B. J., Du, X., Reiter, M. S., & Stewart, R. D. (2020). A calculator to quantify cover crop effects on soil health and productivity. *Soil and Tillage Research*, 199, 104575. <https://doi.org/10.1016/j.still.2020.104575>
- Kiaer, L. P., Skovgaard, I. M., & Ostergaard, H. (2009). Grain yield increase in cereal variety mixtures. A meta-analysis of field trials. *Field Crops Research*, 114, 361–373. <https://doi.org/10.1016/j.fcr.2009.09.006>
- Kim, D. G., Kirschbaum, M. U., & Beedy, T. L. (2016). Carbon sequestration and net emissions of CH₄ and N₂O under agroforestry: Synthesizing available data and suggestions for future studies. *Agriculture, Ecosystems & Environment*, 226, 65–78. <https://doi.org/10.1016/j.agee.2016.04.011>
- Kim, N., Zabaloy, M. C., Guan, K., & Villamil, M. B. (2020). Do cover crops benefit soil microbiome? A meta-analysis of current research. *Soil Biology and Biochemistry*, 142, 107701. <https://doi.org/10.1016/j.soilbio.2019.107701>
- King, A. E., & Blesh, J. (2018). Crop rotations for increased soil carbon: Perenniality as a guiding principle. *Ecological Applications*, 28(1), 249–261. <https://doi.org/10.1002/eap.1648>
- Knapp, S., & van der Heijden, M. G. (2018). A global meta-analysis of yield stability in organic and conservation agriculture. *Nature Communications*, 9(1). <https://doi.org/10.1038/s41467-018-05956-1>
- Kuyah, S., Whitney, C. W., Jonsson, M., Sileshi, G. W., Öborn, I., Muthuri, C. W., & Luedeling, E. (2019). Agroforestry delivers a win-win solution for ecosystem services in sub-Saharan Africa. A meta-analysis. *Agronomy for Sustainable Development*, 39(5). <https://doi.org/10.1007/s13593-019-0589-8>
- Letourneau, D. K., Armbricht, I., Rivera, B. S., Lerma, J. M., Carmona, E. J., Daza, M. C., Escobar, S., Galindo, V., Gutierrez, C., Lopez, S. D., Mejia, J. L., Rangel, A. M. A., Rangel, J. H., Rivera, L., Saavedra, C. A., Torres, A. M., & Trujillo, A. R. (2011). Does plant diversity benefit agroecosystems? A synthetic review. *Ecological Applications*, 21, 9–21. <https://doi.org/10.1890/09-2026.1>
- Li, C., Hoffland, E., Kuypers, T. W., Yu, Y., Zhang, C., Li, H., Zhang, F., & van der Werf, W. (2020). Syndromes of production in intercropping impact yield gains. *Nature Plants*, 6(6), 653–660. <https://doi.org/10.1038/s41477-020-0680-9>
- Lichtenberg, E. M., Kennedy, C. M., Kremen, C., Batáry, P., Berendse, F., Bommarco, R., Bosque-Pérez, N. A., Carvalheiro, L. G., Snyder, W. E., Williams, N. M., Winfree, R., Klatt, B. K., Åström, S., Benjamin, F., Brittain, C., Chaplin-Kramer, R., Clough, Y., Danforth, B., Diekötter, T., ... Crowder, D. W. (2017). A global synthesis of the effects of diversified farming systems on arthropod diversity within fields and across agricultural landscapes. *Global Change Biology*, 23(11), 4946–4957. <https://doi.org/10.1111/gcb.13714>
- Lori, M., Symonczik, S., Mader, P., De Deyn, G., & Gattinger, A. (2017). Organic farming enhances soil microbial abundance and activity—A meta-analysis and meta-regression. *PLoS ONE*, 12. <https://doi.org/10.1371/journal.pone.0180442>
- Lowe, E. B., Groves, R., & Gratton, C. (2021). Impacts of field-edge flower plantings on pollinator conservation and ecosystem service delivery—A meta-analysis. *Agriculture, Ecosystems & Environment*, 310, 107290. <https://doi.org/10.1016/j.agee.2020.107290>
- Ma, Z., Chen, H. Y., Bork, E. W., Carlyle, C. N., & Chang, S. X. (2020). Carbon accumulation in agroforestry systems is affected by tree species diversity, age and regional climate: A global meta-analysis. *Global Ecology and Biogeography*, 29(10), 1817–1828. <https://doi.org/10.1111/geb.13145>
- Ma, Z. L., & Chen, H. Y. H. (2016). Effects of species diversity on fine root productivity in diverse ecosystems. A global meta-analysis. *Global Ecology and Biogeography*, 25, 1387–1396. <https://doi.org/10.1111/geb.12488>
- MacWilliam, S., Parker, D., Marinangeli, C. P., & Trémorin, D. (2018). A meta-analysis approach to examining the greenhouse gas implications of including dry peas (*Pisum sativum* L.) and lentils (*Lens culinaris* M.) in crop rotations in western Canada. *Agricultural Systems*, 166, 101–110. <https://doi.org/10.1016/j.agsy.2018.07.016>
- Mahal, N. K., Castellano, M. J., & Miguez, F. E. (2018). Conservation agriculture practices increase potentially mineralizable nitrogen: A meta-analysis. *Soil Science Society of America Journal*, 82(5), 1270–1278. <https://doi.org/10.2136/sssaj2017.07.0245>
- Marcillo, G. S., & Miguez, F. E. (2017). Corn yield response to winter cover crops. An updated meta-analysis. *Journal of Soil and Water Conservation*, 72, 226–239. <https://doi.org/10.2489/jswc.72.3.226>
- Mayer, P. M., Reynolds, S. K. Jr, McCutchen, M. D., & Canfield, T. J. (2007). Meta-analysis of nitrogen removal in riparian buffers. *Journal of Environmental Quality*, 36(4), 1172–1180. <https://doi.org/10.2134/jeq2006.0462>
- McDaniel, M. D., Tiemann, L. K., & Grandy, A. S. (2014). Does agricultural crop diversity enhance soil microbial biomass and organic matter dynamics? A meta-analysis. *Ecological Applications*, 24, 560–570. <https://doi.org/10.1890/13-0616.1>
- Meyer, N., Bergez, J. E., Constantin, J., & Justes, E. (2019). Cover crops reduce water drainage in temperate climates: A meta-analysis. *Agronomy for Sustainable Development*, 39(1), 1–11. <https://doi.org/10.1007/s13593-018-0546-y>
- Miguez, F. E., & Bollero, G. A. (2005). Review of corn yield response under winter cover cropping systems using meta-analytic methods. *Crop Science*, 45, 2318–2329. <https://doi.org/10.2135/cropsci2005.0014>
- Morugán-Coronado, A., Linares, C., Gómez-López, M. D., Faz, Á., & Zornoza, R. (2020). The impact of intercropping, tillage and fertilizer type on soil and crop yield in fruit orchards under Mediterranean conditions: A meta-analysis of field studies. *Agricultural Systems*, 178, 102736. <https://doi.org/10.1016/j.agsy.2019.102736>
- Muchane, M. N., Sileshi, G. W., Gripenberg, S., Jonsson, M., Pumariño, L., & Barrios, E. (2020). Agroforestry boosts soil health in the humid and sub-humid tropics:

- A meta-analysis. *Agriculture, Ecosystems & Environment*, 295, 106899. <https://doi.org/10.1016/j.agee.2020.106899>
- Muhammad, I., Sainju, U. M., Zhao, F., Khan, A., Ghimire, R., Fu, X., & Wang, J. (2019). Regulation of soil CO₂ and N₂O emissions by cover crops: A meta-analysis. *Soil and Tillage Research*, 192, 103–112. <https://doi.org/10.1016/j.still.2019.04.020>
- Muhammad, I., Wang, J., Sainju, U. M., Zhang, S., Zhao, F., & Khan, A. (2021). Cover cropping enhances soil microbial biomass and affects microbial community structure: A meta-analysis. *Geoderma*, 381, 114696. <https://doi.org/10.1016/j.geoderma.2020.114696>
- Mupepele, A. C., Keller, M., & Dormann, C. F. (2020). European agroforestry is no universal remedy for biodiversity: A time-cumulative meta-analysis. *bioRxiv*.
- Namatshve, T., Cardinael, R., Corbeels, M., & Chikowo, R. (2020). Productivity and biological N₂-fixation in cereal-cowpea intercropping systems in sub-Saharan Africa. A review. *Agronomy for Sustainable Development*, 40(4), 1–12. <https://doi.org/10.1007/s13593-020-00629-0>
- Nichols, V., Martinez-Feria, R., Weisberger, D., Carlson, S., Basso, B., & Basche, A. (2020). Cover crops and weed suppression in the U.S. Midwest: A meta-analysis and modeling study. *Agricultural & Environmental Letters*, 5, e20022. <https://doi.org/10.1002/acl.20022>
- Ospitan, O. A., Dille, J. A., Assefa, Y., Radicetti, E., Ayeni, A., & Knezevic, S. Z. (2019). Impact of cover crop management on level of weed suppression: A meta-analysis. *Crop Science*, 59(3), 833–842. <https://doi.org/10.2135/crops.c2018.09.0589>
- Paul, B. K., Koge, J., Maass, B. L., Notenbaert, A., Peters, M., Groot, J. C., & Tittonell, P. (2020). Tropical forage technologies can deliver multiple benefits in Sub-Saharan Africa. A meta-analysis. *Agronomy for Sustainable Development*, 40(4), 1–17. <https://doi.org/10.1007/s13593-020-00626-3>
- Peng, S., & Chen, H. Y. (2021). Global responses of fine root biomass and traits to plant species mixtures in terrestrial ecosystems. *Global Ecology and Biogeography*, 30(1), 289–304. <https://doi.org/10.1111/geb.13205>
- Powlson, D. S., Stirling, C. M., Thierfelder, C., White, R. P., & Jat, M. L. (2016). Does conservation agriculture deliver climate change mitigation through soil carbon sequestration in tropical agro-ecosystems. *Agriculture Ecosystems & Environment*, 220, 164–174. <https://doi.org/10.1016/j.agee.2016.01.005>
- Prevedello, J. A., Almeida-Gomes, M., & Lindenmayer, D. B. (2018). The importance of scattered trees for biodiversity conservation: A global meta-analysis. *Journal of Applied Ecology*, 55(1), 205–214. <https://doi.org/10.1111/1365-2664.12943>
- Pumarino, L., Sileshi, G. W., Gripenberg, S., Kaartinen, R., Barrios, E., Muchane, M. N., Midega, C., & Jonsson, M. (2015). Effects of agroforestry on pest, disease and weed control. A meta-analysis. *Basic and Applied Ecology*, 16, 573–582. <https://doi.org/10.1016/j.baae.2015.08.006>
- Quemada, M., Baranski, M., Nobel-de Lange, M. N. J., Vallejo, A., & Cooper, J. M. (2013). Meta-analysis of strategies to control nitrate leaching in irrigated agricultural systems and their effects on crop yield. *Agriculture, Ecosystems & Environment*, 174, 1–10. <https://doi.org/10.1016/j.agee.2013.04.018>
- Reiss, E. R., & Drinkwater, L. E. (2018). Cultivar mixtures: A meta-analysis of the effect of intraspecific diversity on crop yield. *Ecological Applications*, 28(1), 62–77. <https://doi.org/10.1002/eap.1629>
- Riiser, N. M., & Hansen, T. R. (2014). The favorability of rice-agroforestry. A meta-analysis on yield and soil parameters. <https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.880.9067&rep=rep1&type=pdf>
- Rivest, D., Paquette, A., Moreno, G., & Messier, C. (2013). A meta-analysis reveals mostly neutral influence of scattered trees on pasture yield along with some contrasted effects depending on functional groups and rainfall conditions. *Agriculture, Ecosystems & Environment*, 165, 74–79. <https://doi.org/10.1016/j.agee.2012.12.010>
- Rodriguez, C., Carlsson, G., Englund, J. E., Flöhr, A., Pelzer, E., Jeuffroy, M. H., Makowski, D., & Jensen, E. S. (2020). Grain legume-cereal intercropping enhances the use of soil-derived and biologically fixed nitrogen in temperate agroecosystems. A meta-analysis. *European Journal of Agronomy*, 118, 126077. <https://doi.org/10.1016/j.eja.2020.126077>
- Rosalien, E. J., Pita, A. V., Maria, J. S., & Rene, G. A. B. (2017). Shaded coffee and cocoa—Double dividend for biodiversity and small-scale farmers. *Ecological Economics*, 140, 136–145. <https://doi.org/10.1016/j.ecolecon.2017.04.019>
- Scheper, J., Holzschuh, A., Kuussaari, M., Potts, S. G., Rundlöf, M., Smith, H. G., & Kleijn, D. (2013). Environmental factors driving the effectiveness of European agri-environmental measures in mitigating pollinator loss—a meta-analysis. *Ecology Letters*, 16(7), 912–920. <https://doi.org/10.1111/ele.12128>
- Shi, L., Feng, W., Xu, J., & Kuzyakov, Y. (2018). Agroforestry systems: Meta-analysis of soil carbon stocks, sequestration processes, and future potentials. *Land Degradation & Development*, 29(11), 3886–3897. <https://doi.org/10.1002/ldr.3136>
- Sileshi, G. W. (2016). The magnitude and spatial extent of influence of *Faidherbia albida* trees on soil properties and primary productivity in drylands. *Journal of Arid Environments*, 132, 1–14. <https://doi.org/10.1016/j.jaridenv.2016.03.002>
- Sileshi, G., Akinnifesi, F. K., Ajayi, O. C., & Place, F. (2008). Meta-analysis of maize yield response to woody and herbaceous legumes in sub-Saharan Africa. *Plant and Soil*, 307, 1–19. <https://doi.org/10.1007/s11104-008-9547-y>
- Slattery, R. A., Ainsworth, E. A., & Ort, D. R. (2013). A meta-analysis of responses of canopy photosynthetic conversion efficiency to environmental factors reveals major causes of yield gap. *Journal of Experimental Botany*, 64(12), 3723–3733. <https://doi.org/10.1093/jxb/ert207>
- Smithson, J. B., & Lenne, J. M. (1996). Varietal mixtures: A viable strategy for sustainable productivity in subsistence agriculture. *Annals of Applied Biology*, 128(1), 127–158. <https://doi.org/10.1111/j.1744-7348.1996.tb07096.x>
- Thapa, R., Poffenbarger, H., Tully, K. L., Ackroyd, V. J., Kramer, M., & Mirsky, S. B. (2018). Biomass production and nitrogen accumulation by hairy vetch_cereal rye mixtures. A meta-analysis. *Agronomy Journal*, 110(4), 1197–1208.
- Thapa, R., Mirsky, S. B., & Tully, K. L. (2018). Cover crops reduce nitrate leaching in agroecosystems: A global meta-analysis. *Journal of Environmental Quality*, 47(6), 1400–1411. <https://doi.org/10.2134/jeq2018.03.0107>
- Toler, H. D., Augé, R. M., Benelli, V., Allen, F. L., & Ashworth, A. J. (2019). Global meta-analysis of cotton yield and weed suppression from cover crops. *Crop Science*, 59(3), 1248–1261. <https://doi.org/10.2135/crops.c2018.10.0603>
- Tonitto, C., David, M. B., & Drinkwater, L. E. (2006). Replacing bare fallows with cover crops in fertilizer-intensive cropping systems. A meta-analysis of crop yield and N dynamics. *Agriculture Ecosystems & Environment*, 112, 58–72. <https://doi.org/10.1016/j.agee.2005.07.003>
- Ugarte, C. M., Kwon, H., Andrews, S. S., & Wander, M. M. (2014). A meta-analysis of soil organic matter response to soil management practices. An approach to evaluate conservation indicators. *Journal of Soil and Water Conservation*, 69, 422–430. <https://doi.org/10.2489/jswc.69.5.422>
- Valkama, E., Lemola, R., Kankanen, H., & Turtola, E. (2015). Meta-analysis of the effects of undersown catch crops on nitrogen leaching loss and grain yields in the Nordic countries. *Agriculture Ecosystems & Environment*, 203, 93–101. <https://doi.org/10.1016/j.agee.2015.01.023>
- Van Buskirk, J., & Willi, Y. (2004). Enhancement of farmland biodiversity within set-aside land. *Conservation Biology*, 18(4), 987–994. <https://doi.org/10.1111/j.1523-1739.2004.00359.x>
- VandenBygaart, A. J., Gregorich, E. G., & Angers, D. A. (2003). Influence of agricultural management on soil organic carbon. A compendium and assessment of Canadian studies. *Canadian Journal of Soil Science*, 83(4), 363–380. <https://doi.org/10.4141/S03-009>
- Venter, Z. S., Jacobs, K., & Hawkins, H. J. (2016). The impact of crop rotation on soil microbial diversity. A meta-analysis. *Pedobiologia*, 59, 215–223. <https://doi.org/10.1016/j.pedobi.2016.04.001>
- Verret, V., Gardarin, A., Pelzer, E., Mediene, S., Makowski, D., & Valantin-Morison, M. (2017). Can legume companion plants control weeds without decreasing crop yield? A meta-analysis. *Field Crops Research*, 204, 158–168. <https://doi.org/10.1016/j.fcr.2017.01.010>
- Wang, C., Lamb, E., Zhang, W., Li, X., Zhao, C., Li, C., & Wu, J. (2020). Meta-analysis shows plant diversity enhances grassland carbon and nitrogen cycles. *Authorea Preprints*.
- Wei, H., Xiang, Y., Liu, Y., & Zhang, J. (2017). Effects of sod cultivation on soil nutrients in orchards across China: A meta-analysis. *Soil and Tillage Research*, 169, 16–24. <https://doi.org/10.1016/j.still.2017.01.009>
- Weisberger, D., Nichols, V., & Liebman, M. (2019). Does diversifying crop rotations suppress weeds? A meta-analysis. *PLoS ONE*, 14(7), e0219847. <https://doi.org/10.1371/journal.pone.0219847>
- Winter, S., Bauer, T., Strauss, P., Kratschmer, S., Paredes, D., Popescu, D., Landa, B., Guzmán, G., Gómez, J. A., Guernion, M., Zaller, J. G., & Batáry, P. (2018). Effects of vegetation management intensity on biodiversity and ecosystem services in vineyards: A meta-analysis. *Journal of Applied Ecology*, 55(5), 2484–2495. <https://doi.org/10.1111/1365-2664.13124>
- Wortman, S. E. (2016). Weedy fallow as an alternative strategy for reducing nitrogen loss from annual cropping systems. *Agronomy for Sustainable Development*, 36(4). <https://doi.org/10.1007/s13593-016-0397-3>
- Xu, Z., Li, C., Zhang, C., Yu, Y., van der Werf, W., & Zhang, F. (2020). Intercropping maize and soybean increases efficiency of land and fertilizer nitrogen use. A meta-analysis. *Field Crops Research*, 246, 107661. <https://doi.org/10.1016/j.fcr.2019.107661>
- Yu, Y., Stomph, T. J., Makowski, D., & van der Werf, W. (2015). Temporal niche differentiation increases the land equivalent ratio of annual intercrops. A meta-analysis. *Field Crops Research*, 184, 133–144. <https://doi.org/10.1016/j.fcr.2015.09.010>

- Zamorano, J., Bartomeus, I., Grez, A. A., & Garibaldi, L. A. (2020). Field margin floral enhancements increase pollinator diversity at the field edge but show no consistent spillover into the crop field: A meta-analysis. *Insect Conservation and Diversity*, 13(6), 519–531. <https://doi.org/10.1111/icad.12454>
- Zhang, C., Dong, Y., Tang, L., Zheng, Y., Makowski, D., Yu, Y., Zhang, F., & van der Werf, W. (2019). Intercropping cereals with faba bean reduces plant disease incidence regardless of fertilizer input; a meta-analysis. *European Journal of Plant Pathology*, 154(4), 931–942. <https://doi.org/10.1007/s10658-019-01711-4>
- Zhang, S., Lehmann, A., Zheng, W., You, Z., & Rillig, M. C. (2019). Arbuscular mycorrhizal fungi increase grain yields: A meta-analysis. *New Phytologist*, 222(1), 543–555. <https://doi.org/10.1111/nph.15570>
- Zhao, J., Yang, Y., Zhang, K., Jeong, J., Zeng, Z., & Zang, H. (2020). Does crop rotation yield more in China? A meta-analysis. *Field Crops Research*, 245, 107659. <https://doi.org/10.1016/j.fcr.2019.107659>

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Beillouin, D., Ben-Ari, T., Malézieux, E., Seufert, V., & Makowski, D. (2021). Positive but variable effects of crop diversification on biodiversity and ecosystem services. *Global Change Biology*, 00, 1–14. <https://doi.org/10.1111/gcb.15747>